

# hw5

chl

中文教材：p48, Ex. 1, 4, 10-13.

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1.

假定生男孩生女孩是等可能的，试证在有 3 个孩子的家庭中，事件

$$E = \{\text{此家庭至多有 1 个女孩}\} \quad F = \{\text{家庭中有男孩也有女孩}\}$$

相互独立。如果考虑有 2 个孩子的家庭呢？

*Proof.*  $E$ : 没有女孩 1 种，恰一个女孩 3 种，共 4 种， $\mathfrak{P}(E) = \frac{1}{2}$ ；

$F$ : 不包括全是男孩和全是女孩，有 6 种， $\mathfrak{P}(F) = \frac{3}{4}$ ；

$EF$ : 恰 1 个女孩，共 3 种， $\mathfrak{P}(EF) = \frac{3}{8}$ 。

于是

$$\mathfrak{P}(EF) = \frac{3}{8} = \frac{1}{2} \cdot \frac{3}{4} = P(E) \cdot P(F).$$

独立。

□

4.

设事件  $A, B, C$  独立，求证  $A \cup B$ ,  $A - B$  都与  $C$  独立。

*Proof.*

$$\begin{aligned}
\mathfrak{P}((A \cup B)C) &= \mathfrak{P}(AC) + \mathfrak{P}(BC) - \mathfrak{P}(ABC) \\
&= \mathfrak{P}(A)\mathfrak{P}(C) + \mathfrak{P}(B)\mathfrak{P}(C) - \mathfrak{P}(A)\mathfrak{P}(B)\mathfrak{P}(C) \\
&= (\mathfrak{P}(A) + \mathfrak{P}(B) - \mathfrak{P}(A)\mathfrak{P}(B))\mathfrak{P}(C) \\
&= \mathfrak{P}(A \cup B)\mathfrak{P}(C).
\end{aligned}$$

$$\mathfrak{P}((A - B)C) = \mathfrak{P}(AB^cC) = \mathfrak{P}(A)\mathfrak{P}(B^c)\mathfrak{P}(C) = \mathfrak{P}(AB^c)\mathfrak{P}(C) = \mathfrak{P}(A - B)\mathfrak{P}(C).$$

□

## 10.

试证事件  $A_1, \dots, A_n$  相互独立的充分必要条件是：对每个  $\hat{A}_k = A_k$  或  $\bar{A}_k$  ( $k = 1, \dots, n$ ) 总有

$$P(\hat{A}_1 \cdots \hat{A}_n) = P(\hat{A}_1) \cdots P(\hat{A}_n)$$

*Proof.*  $n = 2$  时。若 A 与 B 独立，则  $\mathfrak{P}(AB) = \mathfrak{P}(A)\mathfrak{P}(B)$ ，于是

$$\mathfrak{P}(A^cB) = \mathfrak{P}(\Omega B) - \mathfrak{P}(AB) = \mathfrak{P}(B) - \mathfrak{P}(A)\mathfrak{P}(B) = \mathfrak{P}(A^c)\mathfrak{P}(B)$$

同理  $\mathfrak{P}(AB^c) = \mathfrak{P}(A)\mathfrak{P}(B^c)$ ，以及

$$\mathfrak{P}(A^cB^c) = \mathfrak{P}(\Omega B^c) - \mathfrak{P}(AB^c) = \mathfrak{P}(B^c) - \mathfrak{P}(A)\mathfrak{P}(B^c) = \mathfrak{P}(A^c)\mathfrak{P}(B^c)$$

反之， $\mathfrak{P}(\hat{A}\hat{B}) = \mathfrak{P}(\hat{A})\mathfrak{P}(\hat{B})$ ，这蕴含  $\mathfrak{P}(AB) = \mathfrak{P}(A)\mathfrak{P}(B)$ ，即 A 与 B 独立。

假设结论对  $n$  成立, 考虑  $n+1$  的情形。若  $A_1, \dots, A_{n+1}$  相互独立, 则  $A_1, \dots, A_n$  相互独立, 于是

$$\mathfrak{P}(\hat{A}_1 \cdots \hat{A}_n A_{n+1}) = \mathfrak{P}(\hat{A}_1 \cdots \hat{A}_n) \mathfrak{P}(A_{n+1}) = \mathfrak{P}(\hat{A}_1) \cdots \mathfrak{P}(\hat{A}_n) \mathfrak{P}(A_{n+1})$$

$$\begin{aligned} \mathfrak{P}(\hat{A}_1 \cdots \hat{A}_n A_{n+1}^c) &= \mathfrak{P}(\hat{A}_1 \cdots \hat{A}_n \Omega) - \mathfrak{P}(\hat{A}_1 \cdots \hat{A}_n A_{n+1}) \\ &= \mathfrak{P}(\hat{A}_1 \cdots \hat{A}_n) - \mathfrak{P}(\hat{A}_1 \cdots \hat{A}_n) \mathfrak{P}(A_{n+1}) \\ &= \mathfrak{P}(\hat{A}_1 \cdots \hat{A}_n) \mathfrak{P}(A_{n+1}^c) = \mathfrak{P}(\hat{A}_1) \cdots \mathfrak{P}(\hat{A}_n) \mathfrak{P}(A_{n+1}^c) \end{aligned}$$

从而

$$P(\hat{A}_1 \cdots \hat{A}_{n+1}) = P(\hat{A}_1) \cdots P(\hat{A}_{n+1})$$

反之,  $\mathfrak{P}(\hat{A}_1 \cdots \hat{A}_{n+1}) = \mathfrak{P}(\hat{A}_1) \cdots \mathfrak{P}(\hat{A}_{n+1})$ , 这蕴含  $\mathfrak{P}(A_1 \cdots A_{n+1}) = \mathfrak{P}(A_1) \cdots \mathfrak{P}(A_{n+1})$ , 即  $A_1, \dots, A_{n+1}$  相互独立。  $\square$

## 11.

假定  $0 < P(B) < 1$ , 试证明事件  $A$  与  $B$  独立的一个充分必要条件是

$$P(A | B) = P(A | \bar{B})$$

*Proof.*

$$\begin{aligned} \mathfrak{P}(A|B) = \mathfrak{P}(A|B^c) &\iff \frac{\mathfrak{P}(AB)}{\mathfrak{P}(B)} = \frac{\mathfrak{P}(AB^c)}{\mathfrak{P}(B^c)} \\ &\iff \mathfrak{P}(AB)\mathfrak{P}(B^c) = \mathfrak{P}(AB^c)\mathfrak{P}(B) \\ &\iff \mathfrak{P}(AB)(1 - \mathfrak{P}(B)) = (\mathfrak{P}(A) - \mathfrak{P}(AB))\mathfrak{P}(B) \\ &\iff \mathfrak{P}(AB) = \mathfrak{P}(A)\mathfrak{P}(B). \end{aligned}$$

等价于  $A$  与  $B$  独立.

□

## 12.

设事件  $A, B, C$  两两独立, 且满足

$$ABC = \emptyset \quad P(A) = P(B) = P(C) = x.$$

试证  $x$  的最大值为  $\frac{1}{2}$ 。

*Proof.* 由容斥原理,

$$\begin{aligned} 1 &\geq \mathfrak{P}(A \cup B \cup C) = \mathfrak{P}(A) + \mathfrak{P}(B) + \mathfrak{P}(C) - \mathfrak{P}(AB) - \mathfrak{P}(AC) - \mathfrak{P}(BC) + \mathfrak{P}(ABC) \\ &= \mathfrak{P}(A) + \mathfrak{P}(B) + \mathfrak{P}(C) - \mathfrak{P}(A)\mathfrak{P}(B) - \mathfrak{P}(A)\mathfrak{P}(C) - \mathfrak{P}(B)\mathfrak{P}(C) \\ &= 3x - 3x^2 = 3x(1 - x) \quad \text{对 } x \in [0, 1] \text{ 恒成立} \end{aligned}$$

由于  $(AB)(AC) = ABC = \emptyset$ , 故  $x = \mathfrak{P}(A) \geq \mathfrak{P}(AB \cup AC) = \mathfrak{P}(AB) + \mathfrak{P}(AC) = 2x^2$ , 解得  $x \leq \frac{1}{2}$ .

等号成立时,  $A = AB \cup AC$ ,  $B = AB \cup BC$ ,  $C = AC \cup BC$ , 以及  $AB, AC, BC$  两两不交。 □

## 13.

如果事件  $A, B, C$  满足

$$P(AB \mid C) = P(A \mid C) \cdot P(B \mid C),$$

其中  $P(C) > 0$ , 则称事件  $A$  与  $B$  关于事件  $C$  条件独立。试证明

(1) 当  $P(BC) > 0$  时, 上述条件独立性的充分必要条件是

$$P(A | BC) = P(A | C).$$

(2) 举例说明“独立”与“条件独立”两者没有蕴含关系。

*Proof.* (1) 当  $P(BC) > 0$  时,

$$\begin{aligned} \mathfrak{P}(AB|C) &= \mathfrak{P}(A|C)\mathfrak{P}(B|C) \iff \frac{\mathfrak{P}(ABC)}{\mathfrak{P}(C)} = \frac{\mathfrak{P}(AC)}{\mathfrak{P}(C)} \frac{\mathfrak{P}(BC)}{\mathfrak{P}(C)} \\ &\iff \frac{\mathfrak{P}(ABC)}{\mathfrak{P}(BC)} = \frac{\mathfrak{P}(AC)}{\mathfrak{P}(C)} \iff \mathfrak{P}(A|BC) = \mathfrak{P}(A|C). \end{aligned}$$

(2) 设  $\Omega = \{w_k : k \leq 4\}$ ,  $\mathfrak{P}(w_k) = \frac{1}{4}$  for  $k = 1, 2, 3, 4$ .  $A = \{w_1, w_2\}$ ,  $B = \{w_1, w_3\}$ ,  $C = \{w_1, w_4\}$ , 可直接验证  $A$  与  $B$  独立, 而

$$\mathfrak{P}(AB|C) = \frac{1/4}{1/2} = \frac{1}{2} \neq \frac{1/4}{1/2} \cdot \frac{1/4}{1/2} = \mathfrak{P}(A|C) \cdot \mathfrak{P}(B|C).$$

事件  $A$  与  $B$  关于事件  $C$  不条件独立。

设  $\Omega = \{w_k : k \leq 6\}$ ,  $\mathfrak{P}(w_k) = \frac{1}{6}$  for  $1 \leq k \leq 6$ .  $A = \{w_1, w_2, w_3\}$ ,  $B = \{w_1, w_4, w_5\}$ ,  $C = \{w_1, w_2, w_4, w_6\}$ , 则

$$\mathfrak{P}(AB) = \frac{1}{6} \neq \frac{1}{2} \cdot \frac{1}{2} = \mathfrak{P}(A) \cdot \mathfrak{P}(B)$$

事件  $A$  与  $B$  不独立, 而

$$\mathfrak{P}(AB|C) = \frac{1/6}{2/3} = \frac{1}{4} = \frac{1/3}{2/3} \cdot \frac{1/3}{2/3} = \mathfrak{P}(A|C) \cdot \mathfrak{P}(B|C).$$

事件  $A$  与  $B$  关于事件  $C$  条件独立。 □